OMIS 105

Introduction to Database Management Systems

Course Outline & Syllabus Overview

Course Roadmap (10 Weeks)

Instructor: Dr. Mahmoud Parsian

Quarter: Fall 2026

Schedule: 2 sessions per week, 2 hours each (4 hours/week)

Prerequisite: OMIS 30 — Introduction to Programming

Welcome to OMIS 105

- Every app you use — Amazon, Instagram, Uber, your bank — is powered by a **database**.
- This course gives you the skills to **design**, **build**, **query**, and **manage** databases that drive real business decisions.
- By the end of this course,
  - you will think like a **data architect**, and
  - write SQL like a **professional analyst**.

What You Will Build

By the end of this course:

- Real databases
- Business-style queries
- Analytical insights
- A complete mini project

Why Databases Matter

- The large majority of **Fortune 1000** companies rely on relational databases
- **SQL** is consistently one of the most requested technical skills in business analytics job postings
- Every business decision — pricing, inventory, marketing, finance — depends on **data stored in databases**
- Database skills bridge the gap between **business strategy** and **technical execution**

> *"Data is the new oil, but only if you know how to refine it."*

Course Structure: The Journey

We move from:

👉 Basics

→ Querying

→ Design

→ Systems

→ Project

What You Will Learn

By the end of this course, you will be able to:

1. **Design** a normalized relational database from business requirements
2. **Write** complex SQL queries to extract business insights
3. **Optimize** database performance with indexes and query tuning
4. **Manage** data integrity through transactions and constraints
5. **Build** a complete database project from scratch
6. **Communicate** database designs using ER diagrams

Our Tools

Tool	Role	Why This Tool?
DuckDB	Database engine	Zero setup, standard SQL, fast analytics
Python	Programming language	Industry standard, DuckDB integration
Marimo Notebooks	Interactive coding	Run SQL, see results instantly, document work
qStudio	SQL Editing	Run SQL, see results instantly

All tools are **free** and **open source**.

Our Dataset: ShopSmart E-Commerce

Throughout the course, we build a realistic **e-commerce database**:

Table	Description	Rows
`products`	64 items across 8 categories	64
`customers`	Buyers with contact info	40
`orders`	Purchase transactions	200
`order_items`	Line items per order	607
`reviews`	Customer product ratings	150
`suppliers`	Product vendors	10
`shipping`	Delivery tracking	141

The dataset **grows each week**, mirroring real-world complexity.

Course Structure: The Big Picture

Weeks	Theme	Focus
Weeks 1–2	FOUNDATIONS	What are databases? How do they work?
Weeks 3–5	SQL MASTERY	The language of data
Week 6	DESIGN	Building databases the right way
Week 7	PERFORMANCE	Making databases fast
Week 8	TRANSACTIONS	Making databases safe
Week 9	PROJECT	Putting it all together
Week 10	SYNTHESIS	Review, trends, and what's next

Each Week You Receive

Material	Description
Slides	Two Marp slide decks covering concepts and examples
Dataset	CSV files that grow in complexity week over week
Demo Notebooks	Marimo notebooks with live SQL demonstrations
Lab (Student)	Hands-on exercises to practice on your own
Lab (Instructor)	Full solutions for reference and grading

Week-by-Week Roadmap

Week 1

Introduction to Databases

- What is a database?
- File systems vs DBMS
- Why databases matter
- First SQL query

👉 Goal: Build confidence

Week 1 — Database Foundations

What You Will Learn

- What databases are and why they replaced spreadsheets
- Core vocabulary: tables, rows, columns, schemas, data types
- Constraints that enforce data quality (PRIMARY KEY, NOT NULL, CHECK)
- Setting up DuckDB and writing your first SQL queries
- Basic SELECT, WHERE, ORDER BY, LIMIT, and aggregate functions

Week 1 — Business Insight

Why This Matters in the Real World

The cost of bad data: IBM estimates poor data quality costs the U.S. economy **$3.1 trillion per year**. Flat files (spreadsheets) break down when:

- Two employees edit the same file simultaneously
- A typo in "Electronics" creates an invisible data silo
- Someone deletes a row and takes critical information with it

Databases solve these problems with structure, constraints, and controlled access. Understanding this foundation is the difference between a business that *reacts* to data problems and one that *prevents* them.

> *Every data-driven company begins with a well-structured database.*

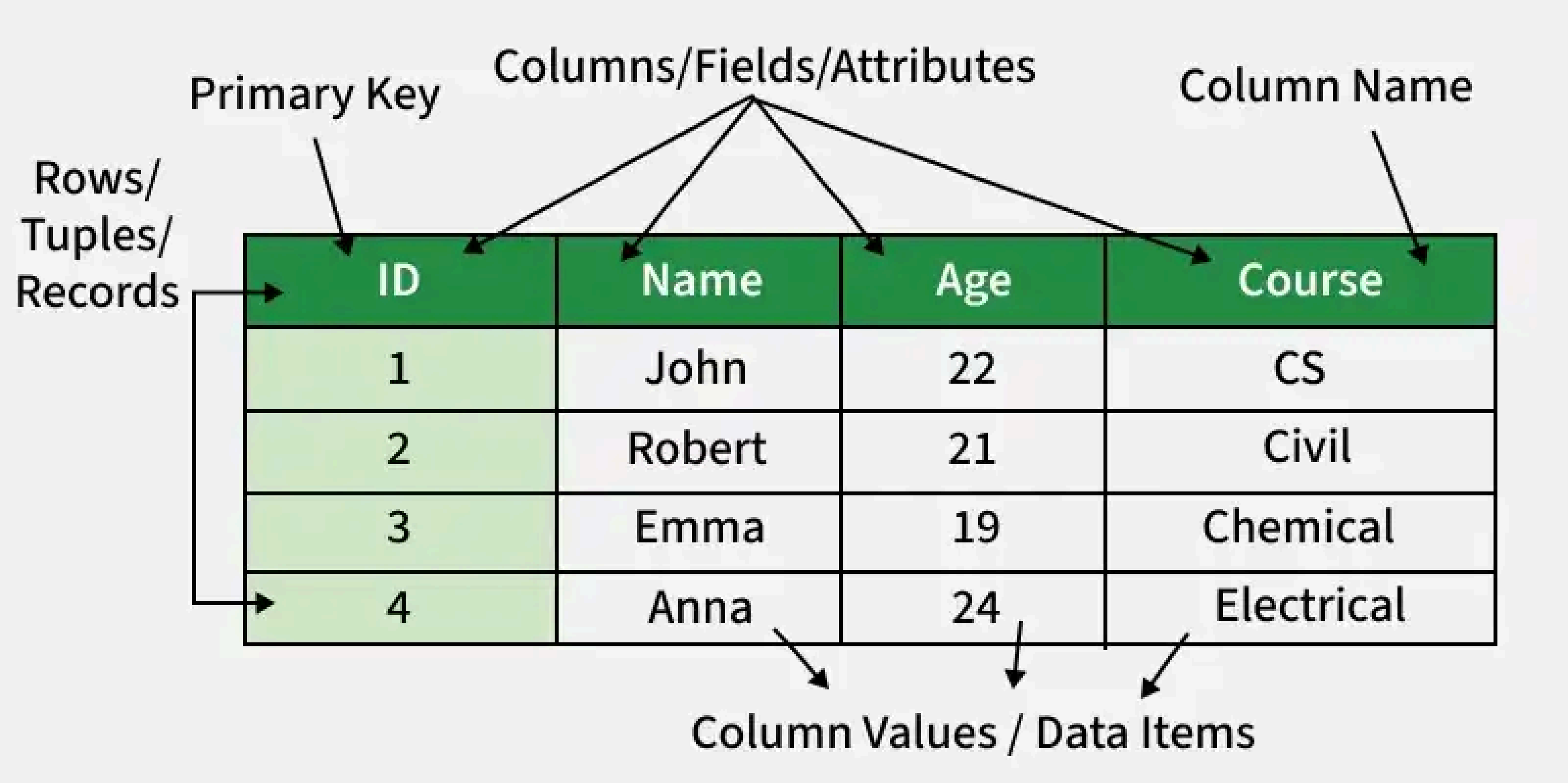
Week 2

Relational Model & Data Modeling

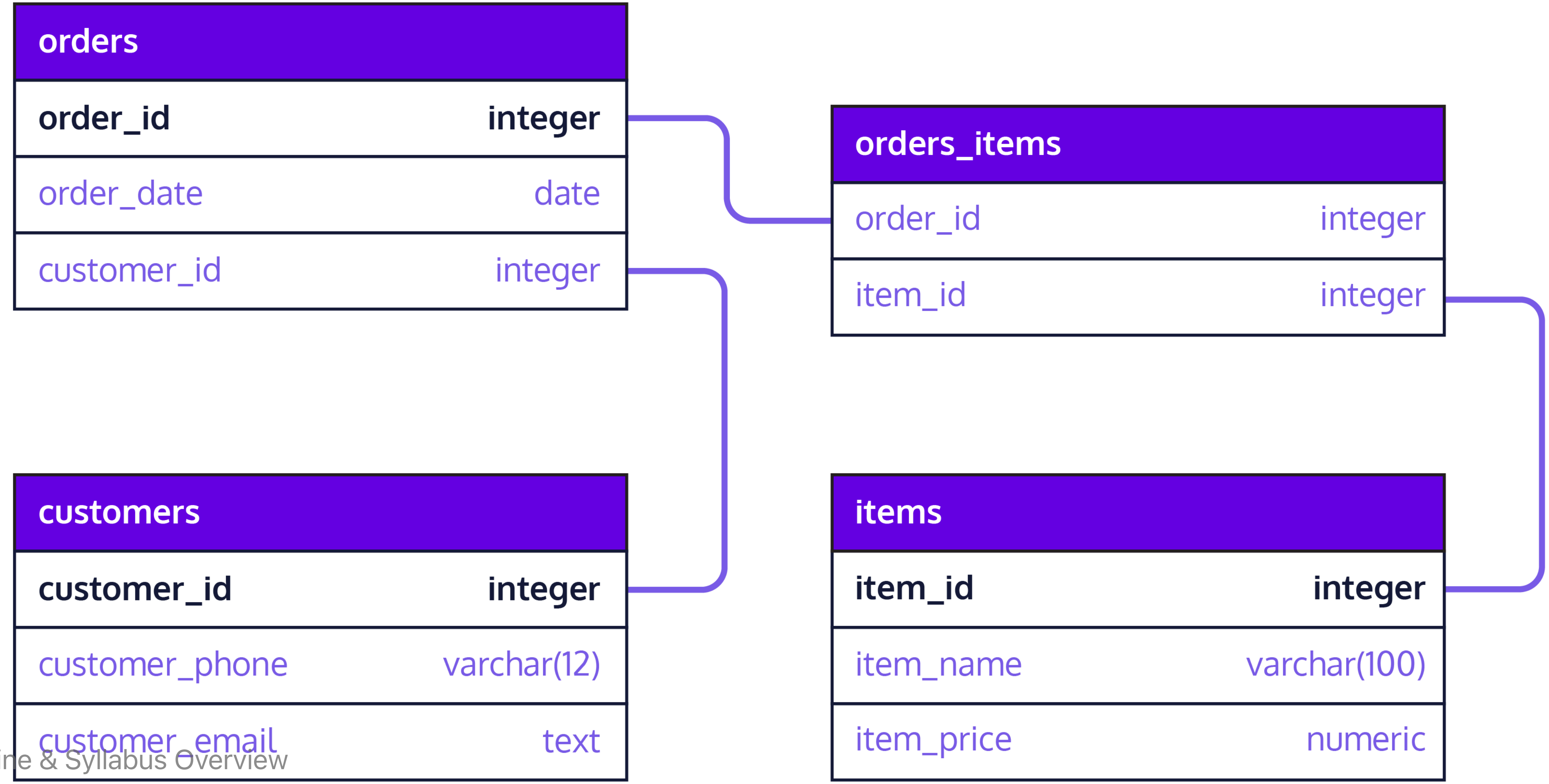
- Tables, rows, columns
- Primary keys, foreign keys
- Relationships (1-1, 1-M, M-M)
  - 1-1 (1 to 1)
  - 1-M (1 to Many)
  - M-M (Many to Many)
- Intro to ER thinking

👉 Goal: Think in structure

Relational Model: A Table: `students`



Relational Model: Set of Tables



Week 2 — Relational Thinking

What You Will Learn

- The relational model (Edgar Codd, 1970 — still dominant today)
- Primary keys, foreign keys, candidate keys, composite keys
- Relationships: one-to-one, one-to-many, many-to-many
- Junction tables for complex relationships
- Entity-Relationship (ER) diagrams using Crow's Foot notation
- Referential integrity — why your data stays consistent

Week 2 — Business Insight

Why This Matters in the Real World

Amazon tracks millions of products, customers, and orders. If customer data were stored inside every order record:

- Updating one email address would require modifying **thousands of rows**
- Deleting a customer's last order would **erase their profile**
- Storage costs would **balloon** with redundant copies

The relational model stores each fact **exactly once** and links tables through keys. This is how businesses maintain **a single source of truth** across terabytes of data.

> *Good relational design is invisible when it works — and catastrophic when it doesn't.*

Week 3

SQL Core (Part 1)

- SELECT
- WHERE
- ORDER BY

👉 Goal: Ask questions with data

Week 3 — SQL Mastery, Part 1: Querying a Single Table

What You Will Learn

- SELECT — choosing which columns come back
- WHERE — filtering rows by a condition
- Comparison and logical operators: `=`, `>`, `<`, `AND`, `OR`, `NOT`
- ORDER BY — sorting results, ascending and descending
- LIMIT — returning only the first N rows
- DISTINCT — removing duplicate values
- Reading a result set critically: is this actually what I asked for?

Week 3 — Business Insight

Why This Matters in the Real World

An inventory manager asks: *"Show me every product over `$700`, most expensive first."*

That is one `SELECT`, one `WHERE`, one `ORDER BY` — and it is the shape of the overwhelming majority of questions asked of a database every day. Before anyone builds a dashboard, somebody has to be able to pull the right rows.

Getting this layer right also builds the habit that matters most: **checking that the rows you got back are the rows you meant to ask for.**

> *Most real SQL is not clever. It is precise.*

Week 4

SQL Core (Part 2)

- Aggregation (`SUM`, `COUNT`, `AVG`, `MIN`, `MAX`)
- `GROUP BY`
- `HAVING`

👉 Goal: Generate insights

Week 4 — SQL Mastery, Part 2: Functions & Aggregation

What You Will Learn

- Aggregate functions: COUNT, SUM, AVG, MIN, MAX
- GROUP BY for summarizing data by category
- HAVING for filtering aggregated groups — and how it differs from WHERE
- String functions: UPPER, LOWER, CONCAT, SUBSTRING, LIKE/ILIKE
- Mathematical functions: ROUND, CEIL, FLOOR, ABS, POWER
- Date functions: EXTRACT, DATEDIFF, CURRENT_DATE, INTERVAL
- Conditional logic with CASE expressions

Week 4 — Business Insight

Why This Matters in the Real World

A marketing manager asks: *"Which product categories have an average price above `$50`, and what percentage of our catalog do they represent?"*

This question requires **grouping** products by category, **aggregating** prices, **filtering** groups by a threshold, and **computing** percentages — all in a single query.

Aggregation is the step where rows become *numbers a manager can act on*. It is what turns raw transactional data into the **executive dashboards** used at companies like Netflix, Spotify, and Target.

> *The ability to ask precise questions of your data is a career-defining skill.*

Week 5

JOINs (Relational Power)

- INNER JOIN
- LEFT JOIN
- RIGHT JOIN
- Multi-table queries

👉 Goal: Connect data

Week 5 — SQL Mastery, Part 3: JOINs

What You Will Learn

- INNER JOIN — combining matching rows from two tables
- LEFT JOIN — keeping all rows from one side (finding gaps)
- RIGHT JOIN and FULL OUTER JOIN
- Self-joins for comparing rows within the same table
- Joining 3, 4, or 5 tables in a single query
- Combining JOINs with GROUP BY and HAVING
- Building multi-table business reports

Week 5 — Business Insight

Why This Matters in the Real World

Real business questions span **multiple tables**:

- *"Who are our top 10 customers by total spending?"* → customers + orders
- *"Which products have never been ordered?"* → products LEFT JOIN order_items
- *"What is revenue by category by month?"* → order_items + products + categories + orders

JOINs are arguably the single most important SQL skill: almost no real dataset lives in one table, so the ability to combine tables correctly is what separates someone who can query a spreadsheet from someone who can query a database.

> *If you master JOINs, you can answer almost any business question.*

Week 6

Database Design & Normalization

- Bad vs good design
- 1NF, 2NF, 3NF (intuitive)
- Reducing redundancy

👉 Goal: Design clean databases

Week 6 — Database Design & Normalization

What You Will Learn

- Functional dependencies: the mathematical foundation of design
- Anomalies: update, insertion, and deletion problems
- First Normal Form (1NF): atomic values, no repeating groups
- Second Normal Form (2NF): eliminating partial dependencies
- Third Normal Form (3NF): eliminating transitive dependencies
- Boyce-Codd Normal Form (BCNF)
- When and how to denormalize for performance
- Hands-on: normalizing a messy denormalized table step by step

Week 6 — Business Insight

Why This Matters in the Real World

Imagine a healthcare system that stores a patient's name inside every appointment record instead of a separate `patients` table. When a patient changes their name after marriage:

- Every appointment row that mentions them needs a manual update, one at a time
- Miss even one, and a claim or a chart now disagrees with the patient's real name
- Nobody finds out until it causes a real problem — an insurance rejection, a mismatched record

Normalization prevents this by ensuring every fact is stored **exactly once**. It is the discipline that separates a database that works for 5 years from one that collapses under its own weight.

> *Normalization is not academic theory — it is insurance against real-world data disasters.*

Week 7

Indexing & Performance

- What is an index?
- Why queries can be slow
- Performance intuition

👉 Goal: Think like a system

Week 7 — Performance & Indexing

What You Will Learn

- How a DBMS executes queries: parsing, optimization, execution
- Full table scans vs. index lookups
- B-Tree indexes: how they work and when to use them
- Composite and unique indexes
- EXPLAIN: reading and interpreting query execution plans
- Six query optimization techniques
- DuckDB's columnar storage and vectorized execution
- Measuring and benchmarking query performance

Week 7 — Business Insight

Why This Matters in the Real World

Imagine a retail company's daily sales report that takes almost an hour to run, because it scans every row of a multi-million-row `orders` table every single time. A DBA adds a few targeted indexes and rewrites the slowest subqueries:

- The same report now finishes in seconds instead of tens of minutes
- The database server has far more headroom to serve other users at the same time
- Nobody needs to buy bigger, more expensive hardware to fix a query that was just poorly written

Performance tuning is not about making things "a little faster." It is about the difference between a system that **scales** and one that **collapses** under load. The techniques in this week directly translate to cost savings and user satisfaction.

> *A slow query is not just a technical problem — it is a business problem.*

Week 8

Transactions & ACID

- Transactions (`BEGIN`, `COMMIT`, `ROLLBACK`)
- ACID properties
- Data correctness

👉 Goal: Understand reliability

Week 8 — Transactions & ACID

What You Will Learn

- What transactions are and why they exist
- ACID properties: Atomicity, Consistency, Isolation, Durability
- BEGIN, COMMIT, ROLLBACK, and SAVEPOINT
- Concurrency problems: dirty reads, lost updates, phantom reads
- Isolation levels: READ UNCOMMITTED through SERIALIZABLE
- Locking mechanisms and deadlock prevention
- Error handling patterns in Python
- Building robust multi-step operations (order processing, returns)

Week 8 — Business Insight

Why This Matters in the Real World

In 2012, Knight Capital Group lost **$440 million in 45 minutes** due to a software deployment that executed trades without proper transaction controls. Partial updates ran unchecked, and there was no rollback mechanism.

Every time a customer clicks "Place Order,"
a transaction must:

1. Create the order record
2. Add each line item
3. Decrement inventory
4. Process payment

Week 8 — Business Insight

Why This Matters in the Real World

- If step 3 fails, steps 1 and 2 must be **undone**.
- ACID guarantees are what make e-commerce, banking, and healthcare systems trustworthy.

> *Transactions are the invisible contract between your system and your users.*

Week 9

Project (Integration)

- Design your own database
- Write real queries
- Generate insights

👉 Goal: Apply everything

Week 9 — Capstone Project

What You Will Do

- Choose a real-world domain (restaurant, gym, hospital, airline, etc.)
- Design an ER diagram with 5+ entities and proper relationships
- Create a normalized schema with constraints
- Load realistic sample data (20+ rows per main table)
- Write 10 diverse SQL queries demonstrating all skills learned
- Build a multi-step transaction with error handling
- Create views and indexes with justification
- Present your work in a live demo (5–8 minutes)

Week 9 — Business Insight

Why This Matters in the Real World

The capstone mirrors what database professionals do every day:

1. A stakeholder describes a business problem
2. You translate requirements into a **data model**
3. You build the schema, load data, and write queries
4. You present findings to **non-technical** decision-makers

This is the workflow at consulting firms (Deloitte, Accenture), tech companies (Google, Meta), and every startup that stores user data. Your capstone project becomes a **portfolio piece** that demonstrates end-to-end database competency.

> *The project is where knowledge becomes capability.*

Week 10

Wrap-up & Modern Data

- Review key concepts
- Common mistakes
- Where SQL is used today

👉 Goal: Big-picture understanding

Week 10 — Synthesis & Review

What You Will Cover

- Capstone project presentations
- Modern database landscape: NoSQL, NewSQL, cloud databases
- Data lakes, lakehouses, and vector databases
- The DBA role: security, backup, capacity planning
- ETL pipelines and data warehousing
- Comprehensive review of all 10 weeks
- Practice problems covering every major topic
- Exam preparation strategies and key patterns to remember

Week 10 — Business Insight

Why This Matters in the Real World

The relational model you have learned is the **foundation** — but the landscape keeps evolving:

- **MongoDB** (document DB) powers real-time apps with flexible schemas
- **Redis** (key-value) handles millions of cache lookups per second
- **Neo4j** (graph DB) maps social networks and fraud detection
- **Snowflake** and **BigQuery** process petabytes in the cloud
- **Pinecone** (vector DB) enables AI-powered search and recommendations

Knowing relational databases gives you the vocabulary to evaluate and adopt any of these technologies. The SQL skills you learned transfer directly to BigQuery, Snowflake, and most modern data platforms.

> *The best database professionals never stop learning — but they all started here.*

Course Policies & Logistics

Grading Breakdown

1000 points total:

Component	Points	Weight	Description
In-Class Labs	600	60%	20 labs × 30 points, one per class session
Midterm Exam	200	20%	Comprehensive, covers Weeks 1–5
Final Exam	200	20%	Comprehensive, covers Weeks 1–10

- Your **lowest lab score is dropped**
- **Midterm and Final Exams**: closed book/notes/internet/software

Weekly Rhythm

Session	When	Structure
Session 1	Day 1, 2 hours	~45 min lecture + live demo, then in-class lab — due by end of session
Session 2	Day 2, 2 hours	~45 min lecture + live demo, then in-class lab — due by end of session

Two labs per week, 20 labs total across the quarter.

Attendance

Each week builds on the previous one — **attendance matters**.

- Each lab must be completed and submitted **during the class session** it's assigned in
- If you are absent, you receive a **zero** on that lab — there is no makeup, since the lab environment and instructor support are only available in class

Academic Integrity

- Labs are **individual** work unless stated otherwise
- You may discuss concepts but must write your own SQL
- Copying queries from classmates or AI tools without understanding them is a violation
- The capstone project must be **your own original design** (not a copy of ShopSmart)
- When in doubt, ask the instructor

Getting Help

- **Office hours**: Announced during the first week of classes
- **Camino Discussions**: Post your question — the instructor, TA, and classmates respond
- **Email**: mparsian@scu.edu
- **Lab sessions:** Bring questions — we work through problems together
- **Peer study groups**: Encouraged! Teaching SQL to others deepens your own understanding

Recommended Preparation

Before Week 1, consider:

1. **Install Python** (3.10+)
2. **Install the tools**: `pip install duckdb pandas marimo`
3. **Review OMIS 30** material — variables, loops, functions
4. **Skim** a SQL tutorial online (any free resource)

None of this is required — we start from the beginning.

The Skills You Will Graduate With

Skill	Outcome
Database Design	Architect data systems
SQL Querying	Extract insights from any dataset
Normalization	Build maintainable, scalable schemas
Performance Tuning	Make systems fast and cost-effective
Transaction Management	Ensure data integrity and safety
Project Delivery	Design, build, present, and defend

These are the skills hiring managers look for in:

Data Analysts, Business Analysts, Software Engineers, Product Managers, and Management Consultants.

How You Will Learn

- Hands-on labs every week
- Real datasets
- Business-style questions
- Step-by-step notebooks

What I Expect From You

- Practice SQL regularly
- Ask questions
- Think in terms of data

What You Can Expect From Me

- Clear explanations
- Practical examples
- Support throughout the course

Final Thought

This course is not about memorizing SQL.

👉 It is about thinking with data

Let's Get Started 🚀

> - *The best time to learn databases was 10 years ago.*
> - *The second best time is today.*

Welcome to OMIS 105.

Let's build something great.

Questions?

Dr. Mahmoud Parsian

mparsian@scu.edu

Leavey School of Business

Santa Clara University

Thank you!

Appendix — Advanced SQL (Optional Enrichment)

Not assessed — beyond the 10-week core

The 10-week core takes SQL as far as **JOINs** in Week 5. The techniques below go further. They appear in several of the `data_stories/` notebooks, so they are listed here for students who want to read ahead — but they are **not taught in the labs and not required for the exam**.

- Window functions: ROW_NUMBER, RANK, LAG, LEAD, NTILE
- Running totals and moving averages
- Percent-of-total calculations
- Common Table Expressions (CTEs) for readable, modular queries
- Set operations: UNION, INTERSECT, EXCEPT
- Creating and using Views — saved queries as virtual tables
- RFM analysis (Recency, Frequency, Monetary) for customer segmentation